

Miguel Benjamin Cervera-Sanchez¹, Daniel San-Juan¹, Roberto Díaz-Peregrino², Evelin Zulema Camacho-Castillo¹,
Sthefany Anahi Bringas-Ortiz³, Christian Padilla-Cabezutd⁴

¹Epilepsy Clinic, National Institute of Neurology and Neurosurgery, Mexico City, Mexico ²Department of Neurosurgery, Heidelberg University Hospital, Ruprecht Karls University of Heidelberg, Heidelberg, Germany ³Autonomous University of Mexico State, Toluca, Mexico ⁴University Center for Health Science, University of Guadalajara, Guadalajara, Mexico

INTRODUCTION

Psychedelics that exert their primary effects through 5-HT_{2A} agonism induce a predictable neural reorganization. Altered serotonergic transmission around epileptic foci has been associated with earlier seizure onset and longer duration, highlighting the need for controlled and selective serotonergic modulation. Psilocybin, a tryptamine alkaloid found in *Psilocybe* mushrooms, and its active metabolite psilocin, induce neurochemical changes primarily through 5-HT_{2A} activation and modulation of cortical neurotransmission. We hypothesize that predominant 5-HT_{2A} modulation may reduce hypersynchrony and cortical hyperexcitability under controlled conditions, but could exacerbate seizures under uncontrolled or high-dose exposure.

METHODS

A narrative literature review was conducted according to the Scale for the Assessment of Narrative Review Articles (SANRA). Literature published from the earliest available year to November 2025 was searched in PubMed/MEDLINE, Web of Science, Scopus, and Google Scholar. The search strategy combined terms related to epilepsy/seizures, psychedelics/psilocybin/psilocin, and serotonergic receptors. Studies evaluating classic psychedelics on serotonergic therapies in seizure contexts, on psilocybin and epilepsy, experimental or clinical studies with specific outcomes, pharmacological studies addressing safety or seizure control.

RESULTS

The search identified 2027 records, of which nine directly relevant studies were included as core studies for synthesis and quality assessment. Additional mechanistic and clinical literature was incorporated to provide context.

Neurobiological and preclinical findings:

Human neuroimaging studies showed that psilocybin induces large-scale brain desynchronization and alters functional connectivity. In animal models, psilocin produced broadband desynchronization and reduced functional connectivity, with 5-HT_{2A} blockade reversing these effects, supporting a central role for 5-HT_{2A} signaling. Experimental seizure models demonstrated that psilocin reduced seizure severity and frequency at selected doses, with involvement of serotonergic, cannabinoid, kynurenine, nitric oxide, and opioid pathways.

Immunological and clinical findings:

Psilocin also demonstrated anti-inflammatory effects in activated macrophages, including reduced TNF- α and increased IL-10, suggesting a potential link between serotonergic modulation and neuroinflammatory mechanisms relevant to epilepsy. Conversely, intracranial neurostimulation documented a marked increase in seizure activity following a high recreational dose of psilocybin-containing mushrooms in a patient with drug-resistant epilepsy, whereas lower subsequent doses did not produce an increase beyond baseline.

Quality assessment:

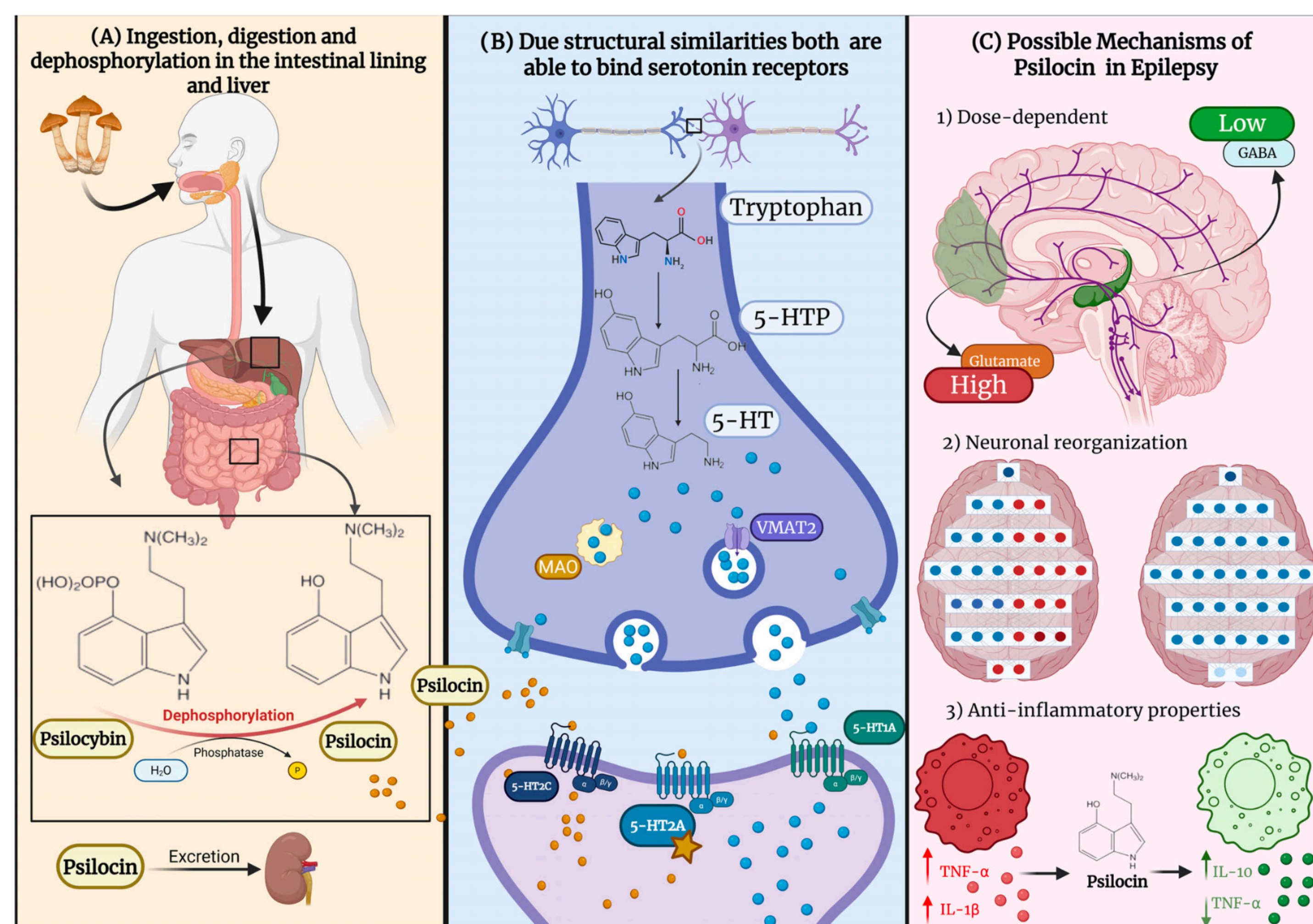
Methodological quality was heterogeneous: recent randomized studies, the case report, and the in vitro study generally demonstrated low RoB, whereas several animal and retrospective studies presented unclear or high RoB.

DISCUSSION

The psilocybin paradox: Current evidence is bidirectional and context-dependent. Preclinical models suggest antiseizure effects, whereas a human case report described seizure exacerbation following high-dose psilocybin exposure in a patient with drug-resistant epilepsy. This apparent discrepancy may reflect an inverted U-shaped dose-response relationship, in which controlled serotonergic activation may be beneficial, while excessive receptor activation or high-dose exposure may promote cortical hyperexcitability. **Serotonergic antiseizure therapies as a benchmark:** The antiseizure effects of fenfluramine, lorcaserin, and other serotonergic agents support serotonin receptors as plausible therapeutic targets in epilepsy. Recent development of selective 5-HT_{2C} agonists, including bexicaserin and BMB-101, further supports the therapeutic potential of subtype-selective serotonergic modulation.

CONCLUSIONS

Psilocybin's ability to modulate 5-HT_{2A}/serotonergic receptors, disrupt pathological neural synchrony, and reorganize functional connectivity offers a potential novel therapeutic paradigm for epilepsy. Evidence remains largely preclinical and highly heterogeneous. While mechanistic overlaps with established serotonergic drugs provide theoretical support, observational reports of proconvulsant activity, warrant extreme caution. Future studies should begin with carefully controlled Phase I/IIa trials, with dose selection, neurological vulnerability, concomitant antiseizure medications, and pharmacokinetic interactions considered as key safety variables to determine whether these theoretical mechanisms can safely translate into clinical practice.



Psilocin: potential therapeutic parallels

